Generalized Linear Models(glm)

13 July 2023 07:12 PM

GLM in R is a class of regression models that supports non-normal distributions and can be implemented in R through glm() function that takes various parameters, and allowing user to apply various regression models like logistic, poission etc

A general linear model makes three assumptions –

Residuals are independent of each other.
Residuals are distributed normally.
Model parameters and y share a linear relationship.

GLM Function

In R, we can use the function glm() to work with generalized linear models in R. Thus, the usage of glm() is like that of the function lm() which we before used for much linear regression. We use an extra argument family. That is to describe the error distribution
And when the model is binomial, the response should be classes with binary values.

glm(formula, family=familytype(link=linkfunction), data=)

1. **Logistic Regression**

We implement the Logistic Regression method for fitting the regression curve y = f(x). Here, y is a categorical variable.

It is a classification algorithm. This model gives out an outcome which is binary in nature. We also use dummy variables to indicate the absence or presence of an effect on the overall result of the model.

The response variable, also known as the dependent variable is categorical in nature. It measures the outcome of the binary response variable. Thus, it actually measures the probability of a binary response.

We use the following R glm() function for modeling our logistic regression method.

glm(response ~ explanantory_variables , family=binomial)

Example

Logistic regression is a simple but powerful model to predict binary outcomes. That is, whether something will happen or not. It's a type of classification model for supervised machine learning.

Logistic regression is used in in almost every industry—marketing, healthcare, social sciences, and others—and is an essential part of any data scientist’s toolkit

1. To predict whether today is going to be a sunny day or not. There are two possible outcomes: "sunny" or "not sunny". The outcome variable is also known as a "target variable" or an "independent variable".
2. Whether a customer will buy your product or not
3. Whether an email is spam or not
4. Whether a transaction is fraudulent or not
5. Whether a drug will cure a patient or not.

There are many variables that could influence the outcome such as ‘temperature the day before’, ‘air pressure’ etc. the influencing variables are known as features, independent variables, or predictors—all these terms mean the same thing.

Logistic Regression implementation

A researcher is interested in how variables, such as GRE (Graduate Record Exam scores), GPA (grade point average) and prestige of the undergraduate institution, affect admission into graduate school. The outcome variable, admit/don’t admit, is binary.

This data set has a binary response (outcome, dependent) variable called admit, which is equal to 1 if the individual was admitted to graduate school, and 0 otherwise. There are three predictor variables: gre, gpa, and rank. We will treat the variables gre and gpa as continuous. The variable rank takes on the values 1 through 4. Institutions with a rank of 1 have the highest prestige, while those with a rank of 4 have the lowest.

```
mydata <- read.csv("C:/Users/user/Downloads/binary.csv")
head(mydata)
str(mydata)

summary(mydata)
xtabs(~admit + rank, data = mydata)
```

```
mydata$rank <- factor(mydata$rank)

mylogit <- glm(admit ~ gre + gpa + rank,data = mydata,family =
"binomial")

new_data = data.frame(gre=380,gpa=3.61,rank=as.factor(3))
new_data2 = data.frame(gre=1000,gpa=4.00,rank=as.factor(1))
summary(mylogit)


predict(mylogit, new_data2, type="response")
```

```
> predict(mylogit, new_data2, type="response")
        1
0.8161678
>
```

Result shows 82 percentage probability for getting admission

2. Poisson Regression

Data is often collected in counts. Hence, many discrete response variables have counted as possible outcomes. While binomial counts are the number of successes in a fixed number of trials, n.

Poisson counts are the number of occurrences of some event in a certain interval of time (or space). Apart from this, Poisson counts have no upper bound and binomial counts only take values between 0 and n.

The basic syntax for glm() function in Poisson regression is −

glm(formula,data,family)

Following is the description of the parameters used in above functions −

formula is the symbol presenting the relationship between the variables.

data is the data set giving the values of these variables.

family is R object to specify the details of the model. It's value is 'Poisson' for Poisson Regression.

Example

We have the in-built data set "warpbreaks" which describes the effect of wool type (A or B) and tension (low, medium or high) on the number of warp breaks per loom. Let's consider "breaks" as the response variable which is a count of number of breaks. The wool "type" and "tension" are taken as predictor variables.

```
input <- warpbreaks
print(head(input))
```

```
output <-glm(formula = breaks ~ wool+tension, data = warpbreaks,
   family = poisson)
print(summary(output))
```

```
Call:
glm(formula = breaks ~ wool + tension, family = poisson, data = warpbreaks)

Deviance Residuals:
    Min       1Q    Median       3Q      Max
  -3.6871  -1.6503  -0.4269    1.1902   4.2616

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.69196    0.04541  81.302  < 2e-16 ***
woolB       -0.20599    0.05157  -3.994 6.49e-05 ***
tensionM    -0.32132    0.06027  -5.332 9.73e-08 ***
tensionH    -0.51849    0.06396  -8.107 5.21e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 297.37  on 53  degrees of freedom
Residual deviance: 210.39  on 50  degrees of freedom
AIC: 493.06

Number of Fisher Scoring iterations: 4
```

In the summary we look for the p-value in the last column to be less than 0.05 to consider an impact of the predictor variable on the response variable. As seen the wooltype B having tension type M and H have impact on the count of breaks.